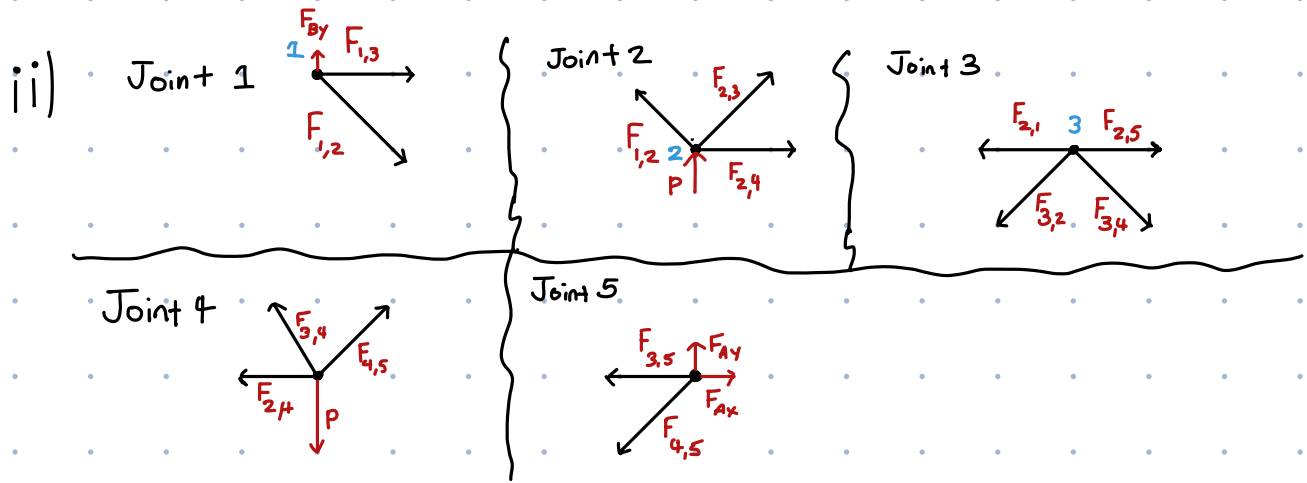
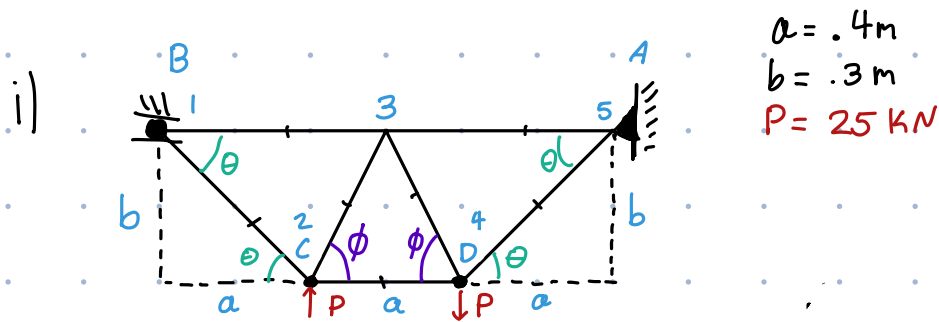




iii)

$$\begin{aligned}
 J_1 \quad \sum F_x &= F_{1,3} + \cos(\theta) F_{1,2} = 0 \\
 \sum F_y &= B_y + \sin(\theta) F_{1,2} = 0 \\
 \sum M &= [A_y \cdot 1.2] + [P \cdot .4] + [-P \cdot .8] = 0
 \end{aligned}
 \left\{ \begin{aligned}
 J_2 \quad \sum F_x &= \cos(\theta) F_{1,2} + \cos(\phi) F_{2,3} + F_{2,4} = 0 \\
 \sum F_y &= \sin(\theta) F_{1,2} + \sin(\phi) F_{2,3} + P = 0
 \end{aligned} \right.$$

$$\begin{aligned}
 J_5 \quad \sum F_x &= A_x + \cos(\theta) F_{4,5} + F_{3,5} = 0 \\
 \sum F_y &= A_y + \sin(\theta) F_{4,5} = 0 \\
 \sum M &= [B_y \cdot 1.2] + [P \cdot .8] + [-P \cdot .4] = 0
 \end{aligned}
 \left\{ \begin{aligned}
 J_4 \quad \sum F_y &= \sin(\phi) F_{3,4} + \sin(\theta) F_{4,5} + P = 0
 \end{aligned} \right.$$

$$\sum M_B = [A_y \cdot 1.2] + [P \cdot 0.4] + [-P \cdot 0.8] = 0$$

$$\sum M_A = [B_y \cdot 1.2] + [P \cdot 0.8] + [-P \cdot 0.4] = 0$$

$$\sum F_x = A_x$$

$$iv) \sum M_B = [A_y \cdot 1.2] + 10 = 20 \quad \sum M_A = 1.2 B_y + 20 - 10 = 0$$

$$A_y = 8.33 \uparrow \quad B_y = 8.33 \downarrow \quad A_x = 0$$

$$\sum F_y = B_y + \sin(\theta) F_{1,2} = 0$$

$$8.33 + \sin(36.87) F_{1,2} = 0$$

$$F_{1,2} = 13.88 \text{ kN (C)}$$

$$\sum F_{y2} = \sin(36.87) \cdot 13.88 + \sin(56.31) F_{2,3} + 25 = 0$$

$$16.67 + \sin(56.31) F_{2,3} = 0$$

$$F_{2,3} = 20.03 \text{ kN (T)}$$

$$\sum F_{x1} = F_{1,3} + \cos(36.87) (-13.88) = 0$$

$$F_{1,3} = 11.10 \text{ kN (T)}$$

$$\sum F_{x2} = -\cos(36.87) F_{1,2} + \cos(56.31) F_{2,3} + F_{2,4} = 0$$

$$-11.10 + 11.10 + F_{2,4} = 0$$

$$F_{2,4} = 0 \text{ kN}$$

$$\sum F_{y5} = -8.33 + \sin(36.87) F_{4,5} = 0$$

$$F_{4,5} = 13.88 \text{ (T) k}$$

$$\sum F_x = \cos(36.87) 13.88 + F_{2,5} = 0$$

$$F_{2,5} = 11.1 \text{ (C) kN}$$

$$\sum F_y = 25 + \sin(56.31) F_{3,4} + \sin(36.87) (13.88) = 0$$

$$F_{3,4} = 20.03 \text{ (C) kN}$$

b) i) Knowns

$$F_{\max} = 20.03 \text{ kN}$$

$$F.S. = 3.5$$

$$\sigma_y = 317 \text{ MPa}$$

$$\rho = 7850 \text{ kg/m}^3$$

Unknowns

$\sigma_{\text{allowed}}$

Area min

Weight

ii)  $\sigma_{\text{allowed}} = \frac{\sigma_y}{F.S.}$  ← Yield Strength

$\sigma_{\text{max}} = \frac{F}{A}$  ← Force

$\sigma_{\text{max}} = \frac{F}{A}$  ← Area

max stress allowed

max stress before snapping

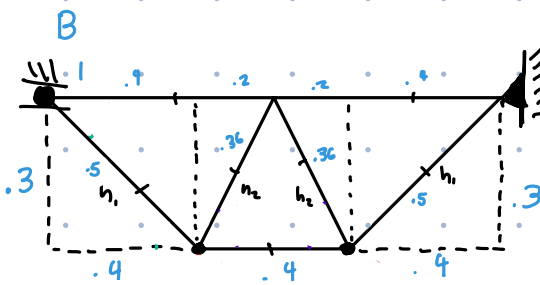
$$\frac{F}{A} = \frac{\sigma_y}{F.S.}$$

$$A_{\min} = \frac{(F.S.) (F)}{\sigma_y}$$

iii)  $A = \frac{(3.5) (20.03 \times 10^3) [\text{N}]}{317 \times 10^6 [\frac{\text{N}}{\text{m}^2}]} = 2.21 \times 10^{-4} \text{ m}^2$

↑ min Area Allowed

iv)



$$h_1 = \sqrt{.4^2 + .3^2} = .5 \text{ m}$$

$$h_2 = \sqrt{.2^2 + .3^2} = .36 \text{ m}$$

$$W = \rho \cdot g \cdot A \cdot L$$

$\frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{m}}{\text{s}^2} \cdot \text{m}^2 \cdot \text{m}$

$$L = 2(.36) + 2(.5) + .4 + 1.2 = 3.32 \text{ m}$$

$$\Rightarrow (7850)(9.81)(2.21 \times 10^{-4})(3.32)$$

$$\Rightarrow 56.5 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \text{ or } N$$

$$\frac{\text{m}^3}{\text{s}^2} \cdot \frac{\text{kg}}{\text{m}^3} \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$\rightarrow 5.76 \text{ kg}$$

3) i)

Known

$$\tau_y = 170 \text{ ksi}$$

$$\rho = .278 \text{ lb/in}^3$$

$$\text{F.S.} = 4$$

$$V_{\text{max}} = 25 \text{ kN}$$

Unknown

Cross Sectional Area

Weight of Pin

ii)



$$V = 25 \text{ kN}$$

or

$$5620.2 \text{ lb}$$

iii)  $\tau_{\text{max}} = \frac{V}{A}$   $\tau_{\text{allow}} = \frac{\tau_y}{\text{F.S.}}$  ← Shear stress

$$\frac{V}{A} = \frac{\tau_y}{\text{F.S.}} \rightarrow A = \frac{(\text{F.S.})(V)}{\tau_y}$$

iv)

$$\Rightarrow A = \frac{(4)(5620.2)}{170 \times 10^3} = .13224 \text{ in}^2$$

$$\frac{\pi d^2}{4} = A$$

$$A_{\text{min}} = 0.13224 \text{ in}^2$$

v)  $W = (\rho)(g)(A)(L)$

$$W = (0.278) \left( \frac{\text{lb}}{\text{in}^3} \right) (386.09) \left( \frac{\text{in}}{\text{s}^2} \right) (.13224) (.732)$$

$$W_{\text{Pin}} = 10.390 \frac{\text{lb} \cdot \text{in}}{\text{s}^2} \text{ or } \text{lbm}$$

$$W = 0.02691 \text{ lbf times 5 Pins}$$

$$W_{\text{tot}} = 0.13455 \text{ lb}$$

for L the Area of Beam of

Truss is  $2.21 \times 10^{-4} \text{ m}^2$ ; Square Cross section

Means side length of  $1.487 \times 10^{-2} \text{ m}$

So for Pin length I just times by 1.25

$$\therefore L = 1.859 \times 10^{-2} \text{ m}$$

$$\downarrow$$

$$0.732 \text{ in}$$

$$g = 32.2 \text{ ft/s}^2 \rightarrow 386.09 \text{ in/s}^2$$

